CLASS 9 — MATHS · FORMULA SHEET

Chapter 3: **World of Numbers** | Quick revision, one page

Rational and irrational numbers

- Rational number: $\frac{p}{q}$ with p, q integers and $q \neq 0$. Irrational number: cannot be written in this form (e.g. $\sqrt{2},\ \sqrt{3},\ \pi$).
- Rational numbers have terminating or non-terminating repeating decimals; irrational numbers have non-terminating, non-repeating decimals.
- $\sqrt{p}$ is irrational for a prime p.

Decimal expansions

- $\frac{p}{q}$ in lowest terms terminates if $q = 2^m 5^n$ $(m, n \geq 0)$; the number of decimal places is the larger of m and n.
- Otherwise the expansion is non-terminating repeating, e.g. $\frac{1}{3} = 0.\overline{3}$.
- Recurring decimal to fraction: for $x = 0.\overline{ab}$, $100x - x = ab$, so $x = \frac{ab}{99}$; for $x = 0.\overline{a}$, $x = \frac{a}{9}$.

Operations on irrationals

- Rational $\pm$ irrational = irrational; non-zero rational $\times$ irrational = irrational.
- Two irrationals can give a rational result: $\sqrt{2} \times \sqrt{8} = 4$, $\sqrt{3} \times \sqrt{3} = 3$.
- $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$, and $(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$ for $a, b > 0$.

Brahmagupta's rules (fortune $+$, debt $-$)

- Fortune $\times$ fortune = fortune; debt $\times$ debt = fortune; fortune $\times$ debt = debt.
- $a - 0 = a$, $a \times 0 = 0$; zero minus zero is zero.

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